

Claude Code on Android via Termux — Runbook

Tested on: Android (ARM64), Termux from F-Droid, Claude Code v2.1.177

Why This Is Non-Trivial

Claude Code ships native binaries for `linux-arm64` and `linux-arm64-musl`, but Android is detected as `linux-arm64-android` — a platform not supported by the installer. The native binaries are also compiled as non-PIE (`ET_EXEC`) which Android's dynamic linker rejects. The workaround is to run Claude Code inside a real Linux environment (Ubuntu) via `proot-distro`.

Prerequisites

- Android phone (ARM64 / 64-bit)
 - **Termux installed from F-Droid** — NOT the Google Play Store version (abandoned)
 - <https://f-droid.org/app/com.termux>
 - A Claude account (Pro, Max, Team, or Enterprise) or Anthropic API key
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Step 1 — Update Termux

```
pkg update && pkg upgrade
```

Step 2 — Install proot-distro and Ubuntu

```
pkg install proot-distro
proot-distro install ubuntu
```

The Ubuntu download takes a few minutes.

Step 3 — Log into Ubuntu

```
proot-distro login ubuntu
```

Your prompt changes to `root@localhost:~#` . All remaining steps run inside here.

Step 4 — Install NVM and Node.js 20

Do **not** use Ubuntu's system Node.js (`apt install nodejs`) — it installs old package versions that conflict with Claude Code's dependencies.

```
apt install -y curl
curl -o- https://raw.githubusercontent.com/nvm-sh/nvm/v0.39.0/install.sh | bash
export NVM_DIR="$HOME/.nvm"
[ -s "$NVM_DIR/nvm.sh" ] && \. "$NVM_DIR/nvm.sh"
nvm install 20
```

NVM automatically adds itself to `~/.bashrc` , so it persists across sessions.

Step 5 — Install Claude Code

```
npm install -g @anthropic-ai/claude-code
```

You should see `Checksums matched!` confirming the native binary downloaded.

Verify:

```
claude --version  
# Expected: 2.1.177 (Claude Code)
```

Step 6 — Authenticate

```
claude
```

When the login URL appears: - Press `c` to copy it (or read it from the screen) - Open it in your Android browser - Log in with your Claude account - Return to Termux — you are now authenticated

Step 7 — Fix PATH and Clean Up (one-time)

Run the built-in diagnostics:

```
/doctor
```

Press `f` to let Claude auto-fix the two issues it finds:

1. `~/local/bin not in PATH` — adds `export PATH="$HOME/.local/bin:$PATH"` to `~/bashrc`
2. **Leftover npm installation** — runs `npm -g uninstall @anthropic-ai/claude-code` to remove the duplicate

After fixes are applied, exit and re-login to pick up the PATH change:

```
exit                # leave Ubuntu
proot-distro login ubuntu # re-enter
which claude        # should show /root/.local/bin/claude
```

Daily Workflow

Every time you want to use Claude Code on your Android:

```
# In Termux:
proot-distro login ubuntu

# Inside Ubuntu:
claude
```

Troubleshooting

Error	Cause	Fix
<code>linux-arm64-android</code> are not available	Android detected instead of Linux	Use proot-distro Ubuntu (this runbook)
<code>has unexpected e_type: 2</code>	Non-PIE binary rejected by Android	Use proot-distro Ubuntu (this runbook)
<code>Cannot find module '.../glob/dist/cjs/src/index.js'</code>	Ubuntu system Node.js too old	Use NVM (Step 4) instead of <code>apt install nodejs</code>
<code>claude: command not found</code> after re-login	NVM not sourced	Run: <code>source ~/.bashrc</code>

Error	Cause	Fix
<code>which claude</code> shows Termux path	Old PATH in current session	Exit and re-login to Ubuntu

What Does NOT Work (and Why)

- **Direct Termux install** (`npm install -g @anthropic-ai/claude-code`): installer detects `linux-arm64-android` , refuses to download binary
- **Force-installing the ARM64 musl binary**: binary is non-PIE (`ET_EXEC`), Android dynamic linker rejects it with `unexpected e_type: 2`
- **Setting `ANTHROPIC_CLAUDE_CODE_PLATFORM` env var**: the install script ignores it
- **Ubuntu system Node.js** (`apt install nodejs`): installs glob v7 which lacks `dist/cjs/src/` structure required by Claude Code